

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course : Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTETC701 - MICROWAVE ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 RF and microwave transmission lines are primarily used to
 - A. Radiate electromagnetic waves
 - B. Guide electromagnetic energy from source to load
 - C. Store electromagnetic energy
 - D. Generate microwave signals
- 2 Time-harmonic electromagnetic waves vary sinusoidally with
 - A. Distance only
 - B. Space only
 - C. Amplitude only
 - D. Time
- 3 A coaxial transmission line supports which mode of propagation?
 - A. TE mode
 - B. TM mode
 - C. TEM mode
 - D. Hybrid mode
- 4 The Y-matrix represents a network in terms of
 - A. Voltages and currents
 - B. Input admittances
 - C. Power and impedance
 - D. Wave amplitudes
- 5 The magic tee combines which types of waveguides?
 - A. E-plane and H-plane
 - B. Rectangular and circular
 - C. Coaxial and microstrip
 - D. TEM and TE modes
- 6 An isolator is a device used to
 - A. Allow power flow in one direction and block in the opposite direction
 - B. Increase transmitted power
 - C. Generate microwaves
 - D. Convert voltage to current
- 7 Microwave tubes are classified as O-type and M-type based on
 - A. Power output
 - B. Electron motion relative to the applied magnetic field
 - C. Operating frequency
 - D. Number of cavities
- 8 In a two-cavity Klystron, the first cavity is called the
 - A. Output cavity
 - B. Coupler cavity
 - C. Reflector cavity
 - D. Buncher cavity
- 9 Microwave power meters are typically used for
 - A. Measuring incident and reflected power
 - B. Measuring electron velocity
 - C. Measuring magnetic field in waveguides
 - D. Calculating cavity dimensions
- 10 Impedance measurement in microwave systems helps in
 - A. Determining line matching to minimize reflections
 - B. Measuring frequency only
 - C. Increasing antenna gain
 - D. Generating standing waves

- 11 Coplanar strip lines are characterized by
- A. Conductors on opposite sides of a dielectric
 - B. Conductors on the same plane with a ground plane nearby
 - C. Helical winding for high frequency
 - D. Shielded coaxial structure
- 12 Radiation hazard levels for personnel are specified in terms of
- A. Current only
 - B. Voltage only
 - C. Power density (W/m^2)
 - D. Impedance

Q2. Solve the following.

- A) Derive the general solution of time-harmonic voltage and current on a lossless transmission line. 6
- B) A rectangular waveguide has dimensions 2.5 X 5 cms. Determine i) guide wavelength ii) phase constant iii) phase velocity at wavelength of 4.5 cms for the dominant mode. 6

Q3. Solve the following.

- A) Explain E-plane, H-plane, and Magic Tee junctions. 6
- B) Explain the ABCD (transmission) matrix representation of two-port networks. 6

Q4. Solve Any Two of the following.

- A) A two cavity klystron amplifier has following parameters $V_0 = 1000V$, $R_0 = 40K\Omega$, $I_0 = 25$ mA, and $f = 3GHz$. Gap spacing for either cavity $d = 1mm$, Spacing between the two cavities $L = 4$ cm, Effective shunt impedance excluding beam loading $= R_{sh} = 30k\Omega$
 - a. Find the input gap voltage to give maximum voltage V_1
 - b. Find the voltage gain, neglecting beam loading in the output cavity
 - c. Find the efficiency of the amplifier, neglecting beam loading
- B) Explain the construction and principle of operation of an 8-cavity cylindrical magnetron. 6
- C) Compare O-type and M-type microwave tubes 6

Q5. Solve Any Two of the following.

- A) Explain microwave power measurement techniques. 6
- B) Define VSWR. Explain the construction and working of a VSWR meter. 6
- C) Define Q factor. Explain the measurement of quality factor (Q) of a cavity resonator. 6

Q6. Solve Any Two of the following.

- A) Explain the working and advantages of a shielded strip line. 6
- B) Explain the radiation hazard levels for personnel and radiation hazard limits. 6
- C) Explain the construction and characteristics of a microstrip line. 6

*** End ***